A decorative graphic on the left side of the slide, consisting of a black crosshair overlaid on a blue square, a red square, and a yellow square.

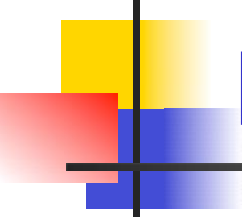
Lecture 9

Chapter 2. Basic Structures

2.2 Set Operations

Set Identities

- Identity: $A \cup \emptyset = A = A \cap U$
- Domination: $A \cup U = U, A \cap \emptyset = \emptyset$
- Idempotent: $A \cup A = A, A \cap A = A$
- Double complement: $\overline{\overline{A}} = A$
- Commutative: $A \cup B = B \cup A, A \cap B = B \cap A$
- Associative: $A \cup (B \cup C) = (A \cup B) \cup C,$
 $A \cap (B \cap C) = (A \cap B) \cap C$
- Distributive: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C),$
 $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- Absorption: $A \cup (A \cap B) = A, A \cap (A \cup B) = A$
- Complement: $A \cup \overline{A} = U, A \cap \overline{A} = \emptyset$

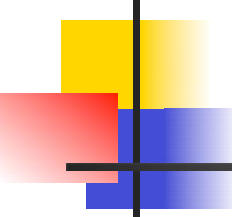
A decorative graphic in the top left corner consisting of overlapping yellow, red, and blue squares with a black crosshair.

DeMorgan's Law for Sets

- Exactly analogous to (and provable from) DeMorgan's Law for propositions.

$$\overline{A \cup B} = \overline{A} \cap \overline{B}$$

$$\overline{A \cap B} = \overline{A} \cup \overline{B}$$

A decorative graphic consisting of overlapping yellow, red, and blue squares with a black crosshair is located in the top left corner.

Proving Set Identities

- To prove statements about sets, of the form $E_1 = E_2$ (where the E s are set expressions), here are three useful techniques:
 1. Prove $E_1 \subseteq E_2$ and $E_2 \subseteq E_1$ separately.
 2. Use set builder notation & logical equivalences.
 3. Use a *membership table*.
 4. Use a Venn diagram.

Method 1: Mutual Subsets

Example: Show $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

■ Part 1: Show $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$.

■ Assume $x \in A \cap (B \cup C)$, & show $x \in (A \cap B) \cup (A \cap C)$.

■ We know that $x \in A$, and either $x \in B$ or $x \in C$.

■ Case 1: $x \in A$ and $x \in B$. Then $x \in A \cap B$,
so $x \in (A \cap B) \cup (A \cap C)$.

■ Case 2: $x \in A$ and $x \in C$. Then $x \in A \cap C$,
so $x \in (A \cap B) \cup (A \cap C)$.

■ Therefore, $x \in (A \cap B) \cup (A \cap C)$.

■ Therefore, $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$.

■ Part 2: Show $(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$. (Try it!)

Method 2: Set Builder Notation & Logical Equivalence



■ Show $\overline{A \cap B} = \bar{A} \cup \bar{B}$

$$\overline{A \cap B} = \{x \mid x \notin (A \cap B)\}$$

def. of complement

$$= \{x \mid \neg(x \in (A \cap B))\}$$

def. of “does not belong”

$$= \{x \mid \neg(x \in A \wedge x \in B)\}$$

def. of intersection

$$= \{x \mid \neg(x \in A) \vee \neg(x \in B)\}$$

De Morgan's law (logic)

$$= \{x \mid x \notin A \vee x \notin B\}$$

def. of “does not belong”

$$= \{x \mid x \in \bar{A} \vee x \in \bar{B}\}$$

def. of complement

$$= \{x \mid x \in \bar{A} \cup \bar{B}\}$$

def. of union

$$= \bar{A} \cup \bar{B}$$

by set builder notation

Method 3: Membership Tables

- Analog to truth tables in propositional logic.
- Columns for different set expressions.
- Rows for all combinations of memberships in constituent sets.
- Use “1” to indicate membership in the derived set, “0” for non-membership.
- Prove equivalence with identical columns.

Membership Table Example

- Prove $(A \cup B) - B = A - B$.

A	B	$A \cup B$	$(A \cup B) - B$	$A - B$
1	1	1	0	0
1	0	1	1	1
0	1	1	0	0
0	0	0	0	0

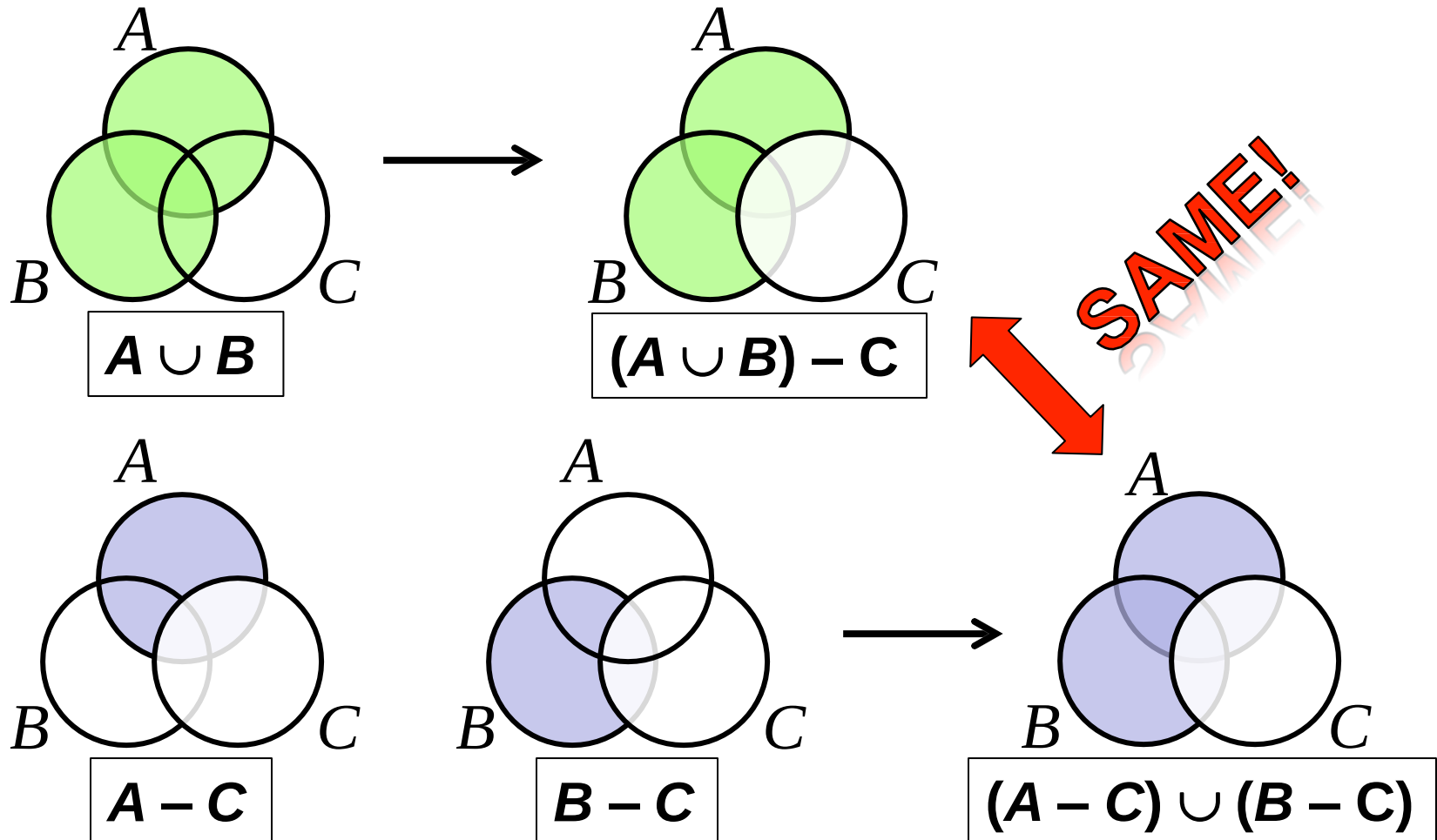
Membership Table Exercise

- Prove $(A \cup B) - C = (A - C) \cup (B - C)$.

A	B	C	$A \cup B$	$(A \cup B) - C$	$A - C$	$B - C$	$(A - C) \cup (B - C)$
1	1	1	1	0	0	0	0
1	1	0	1	1	1	1	1
1	0	1	1	0	0	0	0
1	0	0	1	1	1	0	1
0	1	1	1	0	0	0	0
0	1	0	1	1	0	1	1
0	0	1	0	0	0	0	0
0	0	0	0	0	0	0	0

Method 4: Venn Diagram

- Prove $(A \cup B) - C = (A - C) \cup (B - C)$.



Generalized Unions & Intersections



- Since union & intersection are commutative and associative, we can extend them from operating on pairs of sets A and B to operating on sequences of sets $A_1, \dots, A_n$, or even on sets of sets, $X = \{A \mid P(A)\}$.

Generalized Union

- Binary union operator: $A \cup B$

- n -ary union:

$$A_1 \cup A_2 \cup \dots \cup A_n = (((A_1 \cup A_2) \cup \dots) \cup A_n)$$

(grouping & order is irrelevant)

- “Big U” notation: $\bigcup_{i=1}^n A_i$

- More generally, union of the sets A_i for $i \in I$:

$$\bigcup_{i \in I} A_i$$

- For infinite number of sets: $\bigcup_{i=1}^{\infty} A_i$

Generalized Union Examples

- Let $A_i = \{i, i+1, i+2, \dots\}$. Then,

$$\bigcup_{i=1}^n A_i = A_1 \cup A_2 \cup A_3 \cup \dots \cup A_n$$

$i=1$

$$= \{1, 2, 3, \dots\} \cup \{2, 3, 4, \dots\} \cup \dots \cup \{n, n+1, n+2, \dots\}$$

$$= \{1, 2, 3, \dots\}$$

- Let $A_i = \{1, 2, 3, \dots, i\}$ for $i = 1, 2, 3, \dots$. Then,

$$\bigcup_{i=1}^{\infty} A_i = A_1 \cup A_2 \cup A_3 \cup \dots$$

$i=1$

$$= \{1\} \cup \{1, 2\} \cup \{1, 2, 3\} \cup \dots$$

$$= \{1, 2, 3, \dots\} = \mathbb{Z}^+$$

Generalized Intersection

- Binary intersection operator: $A \cap B$

- n -ary intersection:

$$A_1 \cap A_2 \cap \dots \cap A_n \equiv ((\dots((A_1 \cap A_2) \cap \dots) \cap A_n)$$

(grouping & order is irrelevant)

- “Big Arch” notation: $\bigcap_{i=1}^n A_i$

- Generally, intersection of sets A_i for $i \in I$:

$$\bigcap_{i \in I} A_i$$

- For infinite number of sets:

$$\bigcap_{i=1}^{\infty} A_i$$

Generalized Intersection Examples

- Let $A_i = \{i, i+1, i+2, \dots\}$. Then,

$$\bigcap_{i=1}^n A_i = A_1 \cap A_2 \cap A_3 \cap \dots \cap A_n$$

$i=1$

$$= \{1, 2, 3, \dots\} \cap \{2, 3, 4, \dots\} \cap \dots \cap \{n, n+1, n+2, \dots\}$$

$$= \{n, n+1, n+2, \dots\}$$

- Let $A_i = \{1, 2, 3, \dots, i\}$ for $i = 1, 2, 3, \dots$. Then,

$$\bigcap_{i=1}^{\infty} A_i = A_1 \cap A_2 \cap A_3 \cap \dots$$

$i=1$

$$= \{1\} \cap \{1, 2\} \cap \{1, 2, 3\} \cap \dots$$

$$= \{1\}$$

Bit String Representation of Sets

- A frequent theme of this course are methods of *representing* one discrete structure using another discrete structure of a different type.
- For an enumerable universal set U with ordering $x_1, x_2, x_3, \dots$, we can represent a finite set $S \subseteq U$ as the finite bit string $B = b_1 b_2 \dots b_n$ where $b_i = 1$ if $x_i \in S$ and $b_i = 0$ if $x_i \notin S$.
- E.g. $U = \mathbf{N}$, $S = \{2, 3, 5, 7, 11\}$, $B = 0011\ 0101\ 0001$.
- In this representation, the set operators “ $\cup$ ”, “ $\cap$ ”, “ $-$ ” are implemented directly by bitwise OR, AND, NOT!

Examples of Sets as Bit Strings

- Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, and the ordering of elements of U has the elements in increasing order, then

$$S_1 = \{1, 2, 3, 4, 5\} \Rightarrow B_1 = 11\ 11100000$$

$$S_2 = \{1, 3, 5, 7, 9\} \Rightarrow B_2 = 10\ 10101010$$

- $S_1 \cup S_2 = \{1, 2, 3, 4, 5, 7, 9\}$
 $\Rightarrow$ bit string = 11 1110 1010 = $B_1 \vee B_2$

- $S_1 \cap S_2 = \{1, 3, 5\}$
 $\Rightarrow$ bit string = 10 1010 0000 = $B_1 \wedge B_2$

- $\overline{S_1} = \{6, 7, 8, 9, 10\}$
 $\Rightarrow$ bit string = 00 0001 1111 = $\neg B_1$